

- 6 Many different genes are involved in the production of pigments in mammals. One example is the *TYR* gene.

(a) In humans the *TYR* gene is located on chromosome 11.

Fig. 6.1 shows the homologous pair for chromosome 11.

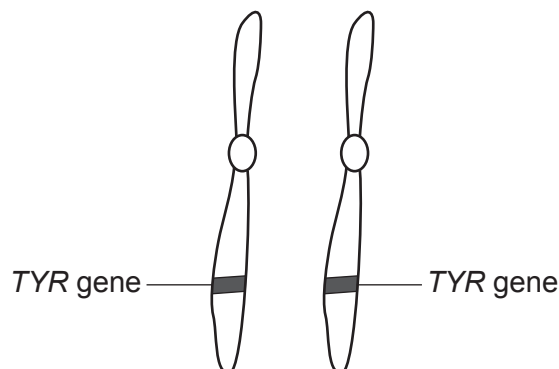


Fig. 6.1

- (i) Homologous chromosomes have the same genes located at the same loci.

State one **other** feature shared by homologous chromosomes.

.....
..... [1]

- (ii) Albinism in humans can be caused by recessive mutations of the *TYR* gene.

Explain why a person who is homozygous recessive for the *TYR* gene shows albinism.

.....
.....
.....
.....
.....
.....
..... [3]

(b) Fur colour in rabbits involves a number of different genes. Some of these genes interact.

Gene 1 has two alleles, **B** and **b**.

- The dominant allele, **B**, results in black fur.
- The recessive allele, **b**, results in brown fur.

Gene 2 has two alleles, **F** and **f**.

- The dominant allele, **F**, codes for a protein that allows the expression of gene 1.
- The recessive allele, **f**, codes for a protein that does not allow the expression of gene 1, resulting in white fur.

The two genes are on different pairs of autosomes.

Complete the genetic diagram for a cross between two black rabbits that are heterozygous for both genes and show the ratio of possible offspring phenotypes.

parental phenotypes black × black

parental genotypes BbFf × BbFf

Punnett square

ratio of offspring phenotypes:

..... [4]

[Total: 8]